



NCCER Module 22107



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Vertical-Mast Sit-Down Counterbalance Forklifts





Section Two

Forklift Fundamentals

Objective

2. Describe the distinguishing features of SDCB forklifts.
 - a. Explain the difference between *OSHA Class 1*, *Class 2*, *Class 4*, and *Class 5* forklifts.
 - b. Identify and describe basic features of vertical mast forklifts.



Performance Task

Under the supervision of an instructor, trainees should be able to do the following:

There are no Performance Tasks associated with this section.



2.0.0 Forklift Fundamentals

- The world's first mass-produced lift was created in 1917 and powered by an internal-combustion (IC) engine.
- The Tructractor transported heavier materials within the factory from one location to another.
- All modern forklifts have a pair of flat tines known as forks, which are used to lift and move materials up and down on a **vertical mast**.



2.1.0 Conventional Forklift Basics

OSHA 1910.178 requires that all forklift operators must be trained.

Forklift operators must understand:

- Proper use
- How a forklift operates
- Forklift controls and instrumentation
- Common hazards
- Load dynamics and balancing
- How and when to do inspections and maintenance



2.1.1 Forklift Uses

- Warehouse forklifts are designed for indoor use but can sometimes be used outdoors, depending on the surface and the type of forklift.
- Conventional forklifts have either pneumatic or hard rubber tires and cleaner emissions than rough-terrain forklifts.



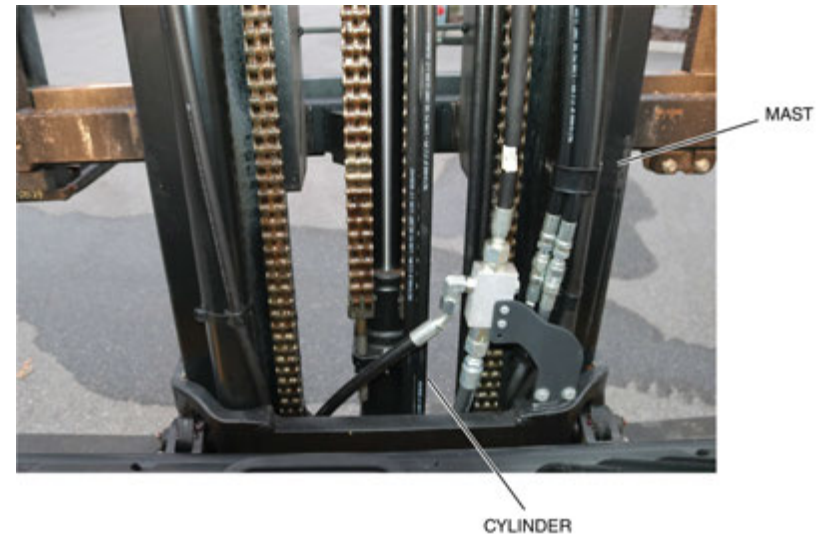
2.1.2 Internal Combustion and Battery Powered

- IC-powered forklifts have an engine mounted low and underneath the operator's seat.
- Battery-powered forklifts position the battery below the seat and far back, which allows the battery's weight to serve as part of the forklift's **counterbalance**.
- IC-powered forklifts are used in open warehouses or well-ventilated areas.
- Battery-powered lifts are typically where proper ventilation or noise may be a concern.



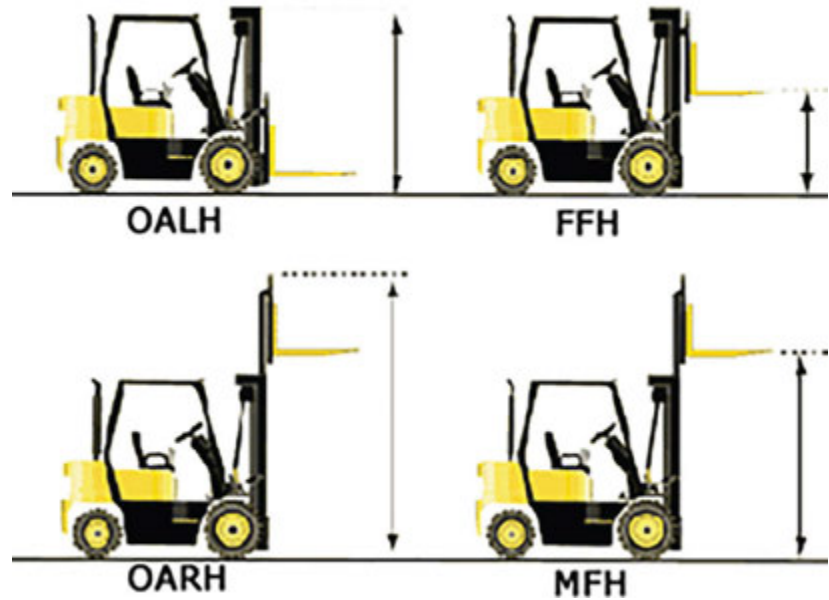
2.2.0 Vertical-Mast Forklifts

- The vertical mast is the upright structure mounted to the chassis that consists of up to four rails.
- The **hydraulic lift cylinder** provides the force that raises the carriage and load.
- To lower the **carriage**, a lever is pushed to open a control valve and allow gravity to force hydraulic fluid back through the return lines and into a reservoir for the hydraulic fluid.



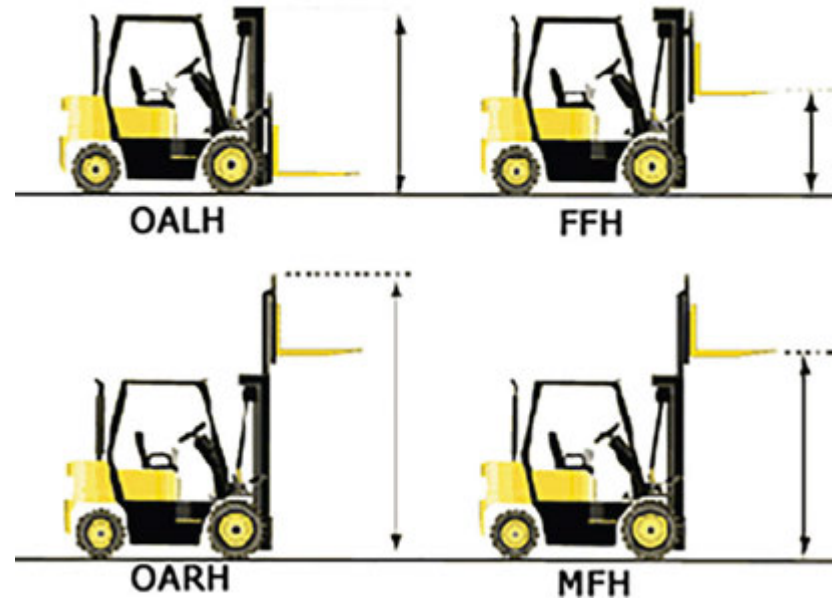
2.2.1 Mast Terminology (1 of 2)

- **Free fork height (FFH)** is the distance the forks can be raised without increasing the overall height of the mast.
- **The overall lowered height (OALH)** is the distance from the ground to the top of the mast when the mast is completely lowered.



2.2.1 Mast Terminology (2 of 2)

- **Maximum fork height (MFH)** is the distance from the ground to the forks when the mast is fully extended.
- **The overall raised height (OARH)** is the distance from the floor to the top of the carriage backrest when the mast is fully extended.
- **Side shift** allows the operator to move the carriage right or left using a hydraulic lever.



2.2.2 Mast Types (1 of 2)

- **Single (Simplex) Stage** – A Simplex mast has one channel with an outer rail and an inner rail that slides up and down. Single stage masts are frequently used to raise loads to low heights when there is little need for ensuring FFH is available. Single stage masts have very little FFH, and low MFH (approximately 48" to 60").
- **Two (Duplex) Stage** – A Duplex mast has two sections and is used in situations where lift height is limited, but considerable FFH (approximately 50") is required. It is perfect for moving loads inside of tractor-trailers. MFH for this type of mast is approximately 88".



2.2.2 Mast Types (2 of 2)

- **Three (Triplex) Stage** – The Triplex (right) has three sections and a primary lift hydraulic cylinder that raises the forks near the top of the mast before the sections begin moving or staging. This mast has considerable FFH and still has an MFH of approximately 189".
- **Four (Quad) Stage** – A Quad (four-section) mast can lift loads up to an MFH of 240".



2.2.3 Classes of Vertical Mast Forklifts (1 of 3)

Class 1

- Powered by an electric motor
- 3 or 4 pneumatic or solid tires
- Used in warehouse applications where air quality and sound are a concern
- Common in storage and retail warehouse environments



2.2.3 Classes of Vertical Mast Forklifts (2 of 3)

Class 2

- Electric
- For use in places that won't accommodate larger lifts

Class 4

- IC engine (LPG, gasoline, or diesel)
- Solid tires and low ground clearance
- Not for use on rough terrain
- Used in larger warehouses and at home and garden retail locations



2.2.3 Classes of Vertical Mast Forklifts (3 of 3)

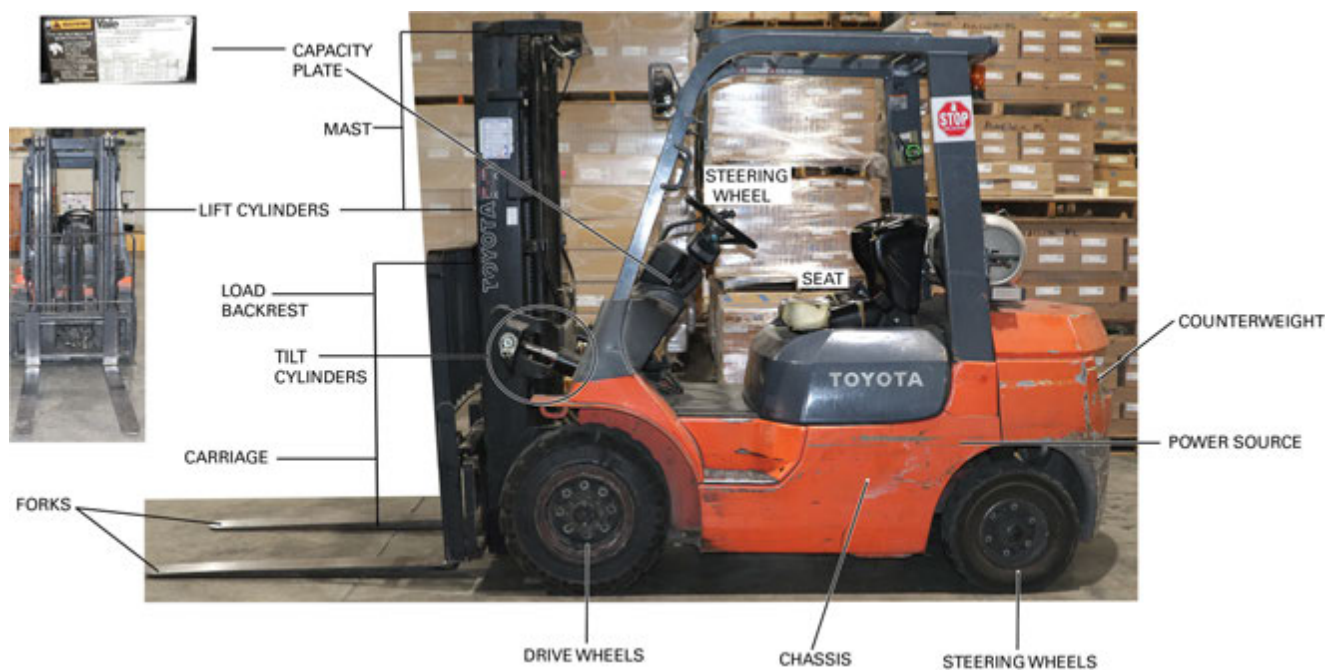
Class 5

- Larger IC engine
- Used for indoor and outdoor applications
- Pneumatic tires
- Limited ground clearance
- Not good for rough-terrain applications



2.2.4 Other Standard Features

Controls and instrumentation may vary depending upon the forklift's class, model, and available options. Read and understand the operator's manual specific to your forklift.



Wrap Up – Trade Terms (1 of 5)

Carriage: Structure on which the forks are mounted. The carriage also serves as the connection point for chains, carriage backrest, and attachments.

Carriage Backrest: An accessory attached to the carriage that serves as a higher backstop for loads lifted with the forks.

Class: A designation for Powered Industrial Trucks (PITs). There are seven PIT classes, but only four of them apply to SDCB forklifts: *OSHA Class 1*, *OSHA Class 2*, *OSHA Class 4*, and *OSHA Class 5*.

Counterbalance: Any weight behind the fulcrum, or front axle, that serves to offset the load, which is any weight in front of the fulcrum.



Wrap Up – Trade Terms (2 of 5)

Counterweight: A weight, usually mounted on the very rear of the forklift, that provides any additional weight needed to counterbalance the lifting capacity.

Falling Object Protective Structure (FOPS): The roof of a ROPS, specifically engineered and designed to protect the forklift's cab and operator from falling items.

Free Fork Height: Also known as Free Lift Height; the distance the forks can be raised without increasing the overall height of the mast.

Hydraulic Lift Cylinder: A single-acting hydraulic cylinder on a SDCB forklift that can only lift the load.



Wrap Up – Trade Terms (3 of 5)

Internal Combustion (IC): A designation given to forklifts that use the combustion of gasoline, diesel, or Liquid Petroleum Gas (LPG) to power the engine.

Lever: A control typically located to the right of the forklift's steering wheel. Conventional forklifts may have between two and four levers. Each lever controls a hydraulic cylinder, which in turn moves the mast, carriage or the load in a specific direction.

Load capacity: The maximum weight the machine can lift based on a load's center, which is usually 2' forward of the carriage backrest and 2' above the top surface of the forks.

Mast Tilt: The feature on SDCB forklifts that tilts the mast; it is normally limited to 20 degrees forward and 10 degrees rearward.



Wrap Up – Trade Terms (4 of 5)

Maximum Fork Height: The distance from the ground to the top of the carrying surface of the forks when the mast is fully extended.

Overall Lowered Height: The distance from the ground to the top of the mast when the mast is completely lowered; the top of the mast cannot get lower than this.

Overall Raised Height: The distance from the floor to the top of the carriage backrest extension when the mast is fully extended; the load on the forks, if it exceeds the carriage backrest, will increase this distance.



Wrap Up – Trade Terms (5 of 5)

Powered Industrial Truck (PIT): A mobile, power-propelled truck used to carry, lift, push, pull, stack, or tier materials. All forklifts belong to a class of PITs, but not all PITs are classified as a forklift. All PITs are regulated under OSHA 1910.178.

Side Shift: A forklift feature that allows the operator to move the carriage and load to the right or left.

Vertical Mast: The vertical structure the carriage is attached to as a load is raised or lowered. The mast is mounted to the chassis and may be comprised of several sections or rails.



2.0.0 Section Review Question 1

1. The _____ is a distinguishing feature of SDCB forklifts.
- a. single-stage hydraulic
 - b. tilting mast
 - c. vertical mast
 - d. carriage side shift



2.0.0 Section Review Question 1 Answer

1. The _____ is a distinguishing feature of SDCB forklifts.
- a. single-stage hydraulic
 - b. tilting mast
 - c. vertical mast**
 - d. carriage side shift



2.0.0 Section Review Question 2

2. What is the primary difference between the different types of masts?
- a. The number of hydraulic cylinders
 - b. Load capacity
 - c. The number of sections
 - d. The size and strength of the chains



2.0.0 Section Review Question 2 Answer

2. What is the primary difference between the different types of masts?
- a. The number of hydraulic cylinders
 - b. Load capacity
 - c. The number of sections**
 - d. The size and strength of the chains



2.0.0 Section Review Question 3

3. _____ are all factors that influence what mast type is chosen for a job.
- a. Forklift class, load capacity, horsepower, and work surface
 - b. Workspace, ceiling height, shelf/rack height, and load size
 - c. Ceiling height, load capacity, landing surface, and load height
 - d. Ceiling height, shelf/rack height, forklift class, and workspace



2.0.0 Section Review Question 3 Answer

3. _____ are all factors that influence what mast type is chosen for a job.
- a. Forklift class, load capacity, horsepower, and work surface
 - b. Workspace, ceiling height, shelf/rack height, and load size**
 - c. Ceiling height, load capacity, landing surface, and load height
 - d. Ceiling height, shelf/rack height, forklift class, and workspace



2.0.0 Section Review Question 4

4. The mast of a vertical-mast forklift can be tilted _____.

- a. 5 degrees rearward
- b. 10 degrees rearward
- c. 15 degrees rearward
- d. 20 degrees rearward



2.0.0 Section Review Question 4 Answer

4. The mast of a vertical-mast forklift can be tilted _____.

- a. 5 degrees rearward
- b. 10 degrees rearward**
- c. 15 degrees rearward
- d. 20 degrees rearward



2.0.0 Section Review Question 5

5. The drive wheels on a conventional forklift are _____.

- a. the smaller set of tires located toward the rear of the forklift
- b. the larger set of tires closest to the mast
- c. the wheels used to steer the forklift
- d. the tires located below the counterweight



2.0.0 Section Review Question 5 Answer

5. The drive wheels on a conventional forklift are _____.

- a. the smaller set of tires located toward the rear of the forklift.
- b. the larger set of tires closest to the mast.**
- c. the wheels used to steer the forklift.
- d. the tires located below the counterweight.



Next...

Section 3.0.0

Instrumentation and Controls



*Read Sections 3.0.0 through 3.3.3 and
complete the Section Review.*